

Intro To Programming Languages

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Programming Languages

- You will need to now many programming languages in cybersecurity , python , php , C++
- For us we will know more about python3 , php

Python



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- Python is simple language that was used to create hacking tools, exploits, and more
- We will know what we are gonna use in cybersecurity only

Python

- Python file ends with .py as
 - File.py , xyz.py
 - And to execute that file and make it work we use
- ```
python3 file.py
```

# Python

---

- But you need to put in the first line

```
#!/usr/bin/python3
```

- (Hint) Remember we put in bash file file.sh the first line

```
#!/bin/bash
```

# Python



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- Lets talk about the first function `print()` that used to print any word or sentence as :

- ----- `file.py` -----

```
#!/usr/bin/python3
```

```
print("this is hossam shady")
```

-----

Then lets execute it as

```
python3 file.py
```

# Python



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- Lets talk about Variables
- Variable is as a box that we can put something inside it and then when we need to open it we will do that for example: file.py

▪ -----

```
#!/usr/bin/python3
```

```
x="hi"
```

```
print(x)
```

-----

```
python3 file.py
```

# Python

---



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- Do you know why it printed "hi" not x

- -----

```
#!/usr/bin/python3
```

```
x="Hossam shady"
```

```
print(x)
```

```
Print("x")
```

-----

```
python3 file.py
```

# Python3



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- When you put numbers in variable you don't need to put it in ""

- -----

```
#!/usr/bin/python3
```

```
y=5
```

```
Z=10
```

```
print(y+z)
```

-----

```
python3 file.py
```

# Python3

---



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- Array in python3 if you need to get the fifth letter of word then use array as :

- -----

```
#!/usr/bin/python3
```

```
name="hossamshady"
```

```
print(name[5])
```

-----

```
python3 file.py
```

# Python3

---



- What if you need to print all letters from the fifth to the end

- -----

```
#!/usr/bin/python3
```

```
name="rednexus"
```

```
print(name[5:])
```

-----

```
python3 file.py
```

# Python3

---



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- What if you need to get the second letter but from the end not from the beginning

- -----

```
#!/usr/bin/python3
```

```
name="redteaming"
```

```
print(name[-2])
```

-----

```
python3 file.py
```

# Python



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- If you need to get from the fifth letter but from the end to the beginning

- -----

```
#!/usr/bin/python3
```

```
name="hossamshady and red nexus"
```

```
print(name[:-5])
```

-----

```
python3 file.py
```



# Python

---

- input() function takes the input from user as :

- -----

```
#!/usr/bin/python3
```

```
x=input("what is your name")
```

```
print(x)
```

-----

Here we will take input from user then store it in x then print it

# Python

---



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- What if you need to put variable inside text

- -----

```
#!/usr/bin/python3
```

```
x="sara"
```

```
print(f"the girl is {x}")
```

```

```

```
python3 file.py
```

# Python



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- We need to get input from user , store it in variable then put it inside text

▪ -----

```
#!/usr/bin/python3
```

```
x=input("what is your name?")
```

```
Print(f"the winner is {x}")
```

-----

```
python3 file.py
```

# Python



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If condition

---

```
#!/usr/bin/python3
```

```
X=10
```

```
if x >= 5:
```

```
 print("ok")
```

```
else:
```

```
 print("failed")
```

# Python



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- If you need to take input from user and then check if it is bigger than 5 or not

- -----

```
#!/usr/bin/python3
```

```
X=input("put a number")
```

```
if x >= 5:
```

```
 print("bigger than 5")
```

```
else:
```

```
 print("smaller than 5")
```

# Python

---



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- if we need to use many conditions use elif

- -----

```
#!/usr/bin/python3
```

```
x=5
```

```
if x == 3:
```

```
 print("x is 3")
```

```
elif x == 4:
```

```
 print("x is 4")
```

```
elif x == 5:
```

```
 print("x is 5")
```

# Python



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- Simple Program

- -----

```
#!/usr/bin/python3
```

```
name=input("what is your name")
```

```
print(f"Dear {name} the service is not available ")
```

```

```

```
python3 file.py
```

# Python



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- To Create function in python we use def
- To make function work we need to call it with
- Let's see -----

```
#!/usr/bin/python3
```

```
def main()
```

```
 Print("this is testing")
```

```
main()
```

# Python

---



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- If you found # or need to put comment we put
- -----  
# this is comment  
# this is another comment and not working

# Python

---



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- To open files with python

- -----

```
#!/usr/bin/python3
```

```
file=open("/home/kali/file.txt","r")
```

```
print(file.read())
```

-----

```
python3 xyz.py
```

# Python

---



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- To write inside file with python
- 

```
file=open("/home/kali/file.txt","w")
file.write("this is new line from python")
file.close()
```

---

```
python3 file.py
```



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# Python

---

- To execute Linux commands with python

▪ -----

```
#!/usr/bin/python3
```

```
import os
```

```
os.system("ls")
```

-----

```
python3 file.py
```



# Python

---

- Small project to scan the target

- -----

```
#!/usr/bin/python3
```

```
import os
```

```
target=input("please enter domain to scan: ")
```

```
os.system(f"nmap {target}")
```

# Python



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- Make python script to see what is the ram and hard disk on device

▪ -----

```
#!/usr/bin/python3
```

```
import os
```

```
os.system("free")
```

```
os.system("df -h")
```

-----

# Python

---



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- Let's use public key and private key to encrypt the data

# Python



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First we make script to generate the keys

---

```
import rsa

publickey, privatekey = rsa.newkeys(512)
print(publickey)
print(privatekey)
```

# Python

---



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- After generating keys lets use public key to encrypt

- -----

```
import rsa
```

```
message="this is red nexus"
```

```
publickey, privatekey = rsa.newkeys(512)
```

```
cipher=rsa.encrypt(message,publickey)
```

```
print(cipher)
```

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- Chat Group
- <https://chat.whatsapp.com/F5KJa2f7CrzBD4dPlCSquA>